Glycolysis and pyruvate oxidation I,II

1- Which of the following correctly identifies the cellular location of glycolysis?

A. Mitochondrial matrix

B. Inner mitochondrial membrane

C. Cytoplasm

D. Nucleus

Answer: C

2- Which of the following enzymes catalyzes the rate-limiting step of glycolysis?

A. Hexokinase

B. Aldolase

C. Pyruvate kinase

D. Phosphofructokinase-1

Answer: D

3- Which of the following correctly describes the preparatory phase of glycolysis?

A. ATP is consumed to phosphorylate glucose and its intermediates

B. ATP is generated exclusively by oxidative phosphorylation

C. Pyruvate is converted into acetyl-CoA

D. Lactate is converted into glucose

Answer: A

4- Which of the following reactions produces NADH during glycolysis?

A. Glucose → Glucose-6-phosphate

B. Fructose-6-phosphate → Fructose-1,6-bisphosphate

C. Glyceraldehyde-3-phosphate → 1,3-bisphosphoglycerate

D. Phosphoenolpyruvate → Pyruvate

Answer: C

5- Which of the following enzymes catalyzes substrate-level phosphorylation during glycolysis?

A. Phosphoglucose isomerase

B. Phosphoglycerate kinase

C. Triose phosphate isomerase

D. Enolase

Answer: B

6- Which of the following correctly describes the role of lactate dehydrogenase under anaerobic conditions?

A. It converts lactate into acetyl-CoA

B. It produces ATP directly from pyruvate

C. It converts pyruvate into acetyl-CoA in the cytoplasm

D. It converts pyruvate into lactate and regenerates NAD^+

Answer: D

7- Which of the following combinations correctly describes the regulation of phosphofructokinase-1 (PFK-1)?

A. AMP activates it, whereas ATP inhibits it

B. ATP activates it, whereas AMP inhibits it

C. Citrate activates it, whereas AMP inhibits it

D. ATP and citrate both activate it

Answer: A

8- Which of the following correctly identifies the coenzymes required by the pyruvate dehydrogenase complex?

A. Biotin, pyridoxal phosphate, and tetrahydrofolate only

B. NADPH, biotin, and vitamin B_{12} only

C. Thiamin pyrophosphate, lipoic acid, CoA, FAD, and NAD^+

D. ATP, GTP, and pyridoxal phosphate only

Answer: C

9- Which of the following explains the effect of fluoride on blood glucose measurement?

A. It activates hexokinase and increases glucose consumption

B. It stimulates lactate dehydrogenase in red blood cells

C. It inhibits pyruvate dehydrogenase in mitochondria

D. It inhibits enolase, helping prevent glycolysis in the collected blood sample

Answer: D

10- Which of the following best explains the effect of arsenate on ATP production during glycolysis?

A. It inhibits hexokinase, preventing glucose phosphorylation

B. It blocks pyruvate formation by inhibiting pyruvate kinase

C. It bypasses the phosphoglycerate kinase reaction, reducing ATP production

D. It prevents the conversion of pyruvate into lactate

Answer: C

11- Which of the following correctly describes the role of the malate–aspartate shuttle in aerobic glycolysis?

A. It transfers the reducing equivalents of cytosolic NADH into mitochondria, allowing an estimated 7 ATP net gain from glycolysis per glucose

B. It transports cytosolic NADH directly across the inner mitochondrial membrane

C. It converts pyruvate into lactate inside mitochondria

D. It produces ATP by directly phosphorylating glucose

Answer: A

12- Which of the following best describes the biological importance of the Rapoport–Luebering cycle in red blood cells?

A. It increases ATP production through oxidative phosphorylation

B. It produces 2,3-bisphosphoglycerate (2,3-BPG), which decreases hemoglobin’s affinity for oxygen

C. It converts glucose into glycogen

D. It generates acetyl-CoA for the Krebs cycle

Answer: B

13- Which of the following best explains the biochemical basis of hemolytic anemia in a patient with pyruvate kinase deficiency?

A. Increased mitochondrial ATP production damages the RBC membrane

B. Excessive NADH production prevents oxygen binding to hemoglobin

C. Reduced ATP production compromises RBC survival, leading to premature destruction

D. Increased acetyl-CoA production inhibits hemoglobin synthesis

Answer: C

14- Which of the following changes would most likely occur in a blood sample left unpreserved at room temperature before glucose analysis?

A. Glucose concentration increases due to ATP hydrolysis

B. Glucose concentration remains unchanged because glycolysis stops outside the body

C. Glucose concentration increases due to lactate oxidation

D. Glucose concentration decreases because blood cells continue to consume glucose through glycolysis

Answer: D

15- Which of the following compounds serves as a feedforward activator of pyruvate kinase in mammalian liver cells?

A. Alanine

B. ATP

C. Fructose-1,6-bisphosphate

D. Citrate

Answer: C